Practical No.:10

Code:

```
~
  GNU nano 8.7.1                                                    sstf.c
#include <stdio.h>
#include <stdlib.h>
#include <stdbool.h>

int main() {
    int requests[] = {98, 183, 41, 122, 14, 124, 65, 67};
    int n = 8;
    int head = 53;
    int total_movement = 0;

    bool visited[8] = {false};

    for (int i = 0; i < n; i++) {
        int min_distance = 9999;
        int index = -1;

        for (int j = 0; j < n; j++) {
            if (!visited[j]) {
                int distance = abs(head - requests[j]);
                if (distance < min_distance) {
                    min_distance = distance;
                    index = j;
                }
            }
        }

        visited[index] = true;
        total_movement += min_distance;
        head = requests[index];
    }

    printf("Total Head Movement = %d\n", total_movement);

    return 0;
}
```

Output:

```
Student@DESKTOP-JJ8CFNB MINGW64 ~
$ nano sstf.c

Student@DESKTOP-JJ8CFNB MINGW64 ~
$ gcc sstf.c -o sstf
./sstf
Total Head Movement = 323

Student@DESKTOP-JJ8CFNB MINGW64 ~
```